

FRAUDLENS AI — GOOGLE VEO MASTER CINEMATIC PRODUCTION SCRIPTS

Complete 6-Part Video Series (Natural Tanglish AI Expert Narration)



VIDEO 1: What FraudLens AI Is + Past & Real-Time Transactions

Target Duration: ~3 Minutes 30 Seconds

Theme: High-Tech Cyber-Fintech Command Center & Real-Time Transaction Scoring Engine

Scene 1.1: The FinTech Dilemma & Platform Hook

- **Duration:** 25 Seconds
- **What Appears on Screen:**

Cinematic opening shot of a futuristic digital banking network with millions of glowing data packets traveling across global financial corridors. Sudden red pulse alerts highlight fraudulent transactions hiding within normal banking traffic. A camera push-in reveals the **FraudLens AI Cyber Shield & Login Screen** on a curved OLED glass display.
- **Natural AI Narration (Tanglish):**

"Machi, oru second yosichu paaru — every single second ulagam full-ah crores of digital transactions nadakithu. Google Pay, card swipe, crypto transfer-nu ellame fraction of a second-la mudiyuthu. But the biggest problem enna theriyuma? Ithula genuine customer transaction ethu, international hacker-oda fraud attack ethu-nu epdi kandupidikkurathu? Normal rule-based system-la pota genuine payments block aagum, illana fraud silent-ah ulla poirum. Athuku than namma create pannirukkom **FraudLens AI** — oru state-of-the-art Explainable AI Financial Fraud & Risk Detection Platform."

- **Technical Process Being Explained:**

The Financial Fraud Problem & Rule Engine Failure: Traditional banking rules (`if amount > 50000: block`) suffer from extreme False Positive Rates (FPR), blocking legitimate users while failing against dynamic botnets. FraudLens AI replaces static heuristics with an adaptive, multi-stage machine learning inference pipeline operating at sub-10ms latencies.

- **Exact Project UI to Show:**

LoginCard with the bioluminescent cyber shield, glowing cyan corner brackets, and 1-Click Role Portals (Admin, Monisha, Mohana, Sowmiya).

- **On-Screen Text:**

FRAUDLENS AI // REAL-TIME RISK INTELLIGENCE

Sub-10ms Inference • Dual-Metric Telemetry • Zero-Leak Security

- **Cinematic Veo Visual Prompt:**

Cinematic macro shot of an illuminated futuristic cyber-financial terminal, sleek frosted glass UI with glowing cyan and soft ivory fiber-optic traces, holographic data nodes pulsing with real-time financial telemetry, depth of field, photorealistic 8k octane render.

- **Transition:** Smooth camera tilt and optical data zoom into the Transactions View.

Scene 1.2: Past Historical Data vs. Live Pre-Auth Interception

- **Duration:** 35 Seconds

- **What Appears on Screen:**

Split-screen composition. On the left: **All Transactions Table** displaying historical transactions with search, pagination, and color-coded risk tags (**LOW - 3%**, **MEDIUM - 68%**, **HIGH - 94%**). On the right: **Payment Gateway (Pre-Auth)** screen showing an incoming real-time payment being captured *before* funds leave the bank account.

- **Natural AI Narration (Tanglish):**

"Okay, ippo namma application-kulla poyitom. First namma purinjikka vendiyathu — transactions rendu type-ah ulla varum. Onnu **Historical Transactions** — athavathu already nadanthu mudinja data. Database-la irukura past transaction history-ah analyze panni, customer baseline behaviour enna-nu system padikkum.

Rendaavathu thaana romba critical: **Real-Time Pre-Auth Evaluation**. Oru customer checkout page-la 'Pay' button click pannumpothu, payment gateway funds-ah release panrathukku munnadiye, FraudLens AI-kitta request anupum: 'Idhu safe-ah, illana fraud-ah?' Namma backend just 4 milliseconds-la decide panni reply anupum!"

- **Technical Process Being Explained:**

Pre-Auth Risk Engine Architecture: The difference between batch forensic

processing and inline pre-authorization risk scoring. FraudLens AI operates as an inline webhook filter evaluating payloads before database commit or card network clearance.

- **Exact Project UI to Show:**

TransactionsView with the searchable table, filtering badges, and the PaymentView Pre-Auth simulation portal.

- **On-Screen Text:**

PRE-AUTH INLINE INTERCEPTION: < 4.2ms

Decision Engine: ALLOW | STEP-UP OTP | HARD BLOCK

- **Cinematic Veo Visual Prompt:**

Dynamic split screen: left side displays an elegant frosted glass data table with glowing emerald and crimson telemetry tags; right side shows an interactive payment terminal interface pulsing with live cryptographic verification beams, hyper-clean fintech aesthetic.

- **Transition:** Focus snap onto an incoming live transaction packet entering the system.

Scene 1.3: What Happens Inside — The Inference Pipeline

- **Duration:** 50 Seconds

- **What Appears on Screen:**

A 3D glass schematic visualizing data flow: A JSON transaction packet enters from the frontend, gets validated by Pydantic, passes through the Feature Engineer (calculating velocity and distance leap), feeds into the XGBoost Champion model, outputs a probability, generates the Section 31 Risk Score, and passes through TreeSHAP.

- **Natural AI Narration (Tanglish):**

"Ippo un friend kitta explain pannu: 'Screen-la click pannathum background-la actual-ah enna nadakkuthu?'

Step 1: User click pannathum, Frontend oru **POST request**-ah

`/api/v1/transactions/evaluate` endpoint-ku anupum.

Step 2: Backend-la FastAPI irukura **Pydantic Schema** antha data-va validate pannum — amount positive-ah irukka, device ID correct-ah irukka-nu.

Step 3: Namma **Feature Engineering Service** run aagum. Customer-oda last 1 hour velocity evvalavu, regular city-la irunthu evvalavu km thalli irukanga-nu extra mathematical features create pannum.

Step 4: Intha features namma **XGBoost Machine Learning model**-kulla pogum. Model instantaneous-ah **Fraud Probability** calculate pannum — for example 91.8%.

Step 5: Athey nerathula, namma custom rule engine separate-ah operational **Risk Score** (94/100) assign pannum. Rendum match aagi, result fraction of a second-la UI-ku return aagum!"

- **Technical Process Being Explained:**

Inference Pipeline Workflow: User Action -> Frontend -> Pydantic Schema Validation -> Feature Extraction -> XGBoost Model Prediction -> Risk Scoring Engine -> TreeSHAP Attribution -> JSON Response -> UI Update.

- **Exact Project UI to Show:**

Live Evaluation Form inside TransactionsView displaying real-time calculation progress indicators.

- **On-Screen Text:**

DATA FLOW: User Click → FastAPI → Feature Engine → XGBoost → TreeSHAP → UI Response

- **Cinematic Veo Visual Prompt:**

Volumetric 3D schematic of streaming digital financial data entering a crystalline holographic AI processing core, glowing circuit paths light up in sequential sapphire and emerald waves, technical telemetry diagnostics overlay.

- **Transition:** Camera pans right to reveal the TreeSHAP feature waterfall diagram.

Scene 1.4: Dual Metric & Explainability (Section 31 Compliance)

- **Duration:** 40 Seconds

- **What Appears on Screen:**

Side-by-side comparison of **ML Fraud Probability (91.8%)** vs **Independent Risk Score (94/100)** in ChatMessageCards. Below it, the **TreeSHAP Waterfall Card** shows directional red/green feature bars pushing the score from baseline 0.12 to 0.94.

- **Natural AI Narration (Tanglish):**

"Inga oru super interesting interview point irukku, kavanama kelu! Namma project-la Section 31 Compliance-nu oru strict standard maintain panrom. Athu enna theriyuma? **Fraud Probability** vera, **Risk Score** vera!"

Probability-na, machine learning model past patterns-ah vachu solra statistical percentage — '91.8% chance idhu fraud-ah irukkalam'. But Risk Score-na, bank regulations, customer balance, velocity, location delta ellam serthu namma system assign panra operational index (94 out of 100).

Athoda kooda, namma **TreeSHAP** use panrom. SHAP-na Shapley Additive Explanations. Machine Learning-a black box-ah vidaama, 'Intha transaction-ku distance 8,420 km jump aanathala +0.44 risk yeruchu, familiar iPhone device-la pannathala -0.21 risk kuranjuchu'-nu crystal clear-ah visual chart-la kaatum!"

- **Technical Process Being Explained:**

Section 31 Dual-Metric Mandate & TreeSHAP: Regulatory compliance demands that model probability does not equal legal causality. SHAP provides local additive feature attributions proving exactly why an auto-block occurred for auditability.

- **Exact Project UI to Show:**

RiskScoreCard and ShapWaterfallCard with live interactive values.

- **On-Screen Text:**

SECTION 31 COMPLIANCE: Fraud Probability $\neq$ Risk Score

TreeSHAP Attribution: distance_from_home (+0.44) | amt_deviation (+0.39)

- **Cinematic Veo Visual Prompt:**

Close-up on a transparent glass interface showcasing horizontal glowing attribution bars, red bars push right indicating risk spikes, green bars pull left indicating trust factors, sharp typography, ambient cool white illumination.

- **Transition:** Smooth fade-out with a rotating cyber emblem into Video 2.

VIDEO 2: Admin Module (Platform Command, Model Lab & Governance)

Target Duration: ~3 Minutes 30 Seconds

Theme: Central Authority, Multi-Model Arena, Automated Drift Detection & Audit Logs

Scene 2.1: Admin Command Dashboard & Executive Metrics

- **Duration:** 35 Seconds
- **What Appears on Screen:**

The full **Command Dashboard** rendering in full glory. Key Performance Metric cards glow at the top: Total Transactions (\$24.8M), Model Precision (99.4%), Blocked Fraud Volume (\$1.82M), Average Inference Latency (3.8ms). Interactive Recharts bar charts show hourly fraud trends.
- **Natural AI Narration (Tanglish):**

"Ippo namma paaka porathu FraudLens AI-oda heart and brain — **Admin Module**. Admin login panna udane, avangalukku complete platform authority kedaikkum. Inga screen-la paatha theriyum: Total transactions evvalavu process aachu, evvalavu fraud attacks-ah namma system block pannirukku, and active ML model evvalavu precision-la run aaguthu-nu executive dashboard live-ah update aagite irukkum. Inga top right-la paaru — 'Backend Connected (57ms)' indicator. Every 25 seconds-ku namma frontend backend health endpoint-ah probe panni system latency-ah monitor pannitte irukkum."

- **Technical Process Being Explained:**

Executive Telemetry Aggregation: The dashboard aggregates real-time records from PostgreSQL/SQLite using optimized indexing and SQL window functions, caching rolling statistical summaries to deliver sub-second rendering.
- **Exact Project UI to Show:**

DashboardView metric hero cards, live alerts section, and transaction volume charts.

- **On-Screen Text:**

ADMIN COMMAND DASHBOARD // SOC LEVEL 3 ACCESS

Live Precision: 99.4% • Latency: 3.8ms • Zero-Downtime Telemetry

- **Cinematic Veo Visual Prompt:**

Sweeping high-angle camera motion across an ultra-modern cyber SOC command wall, multiple high-resolution data visualization widgets displaying glowing financial risk metrics and vector charts, deep blue atmosphere with crisp white illumination.

- **Transition:** Camera zooms into the navigation bar clicking on **Model Lab & Registry**.

Scene 2.2: Model Lab — The Multi-Model Arena

- **Duration:** 45 Seconds

- **What Appears on Screen:**

The **Model Lab & Registry** view. A comparison matrix compares 5 trained algorithms: **XGBoost (Active Champion)**, **Random Forest**, **Decision Tree**, **Logistic Regression**, and **Ensemble Voting Classifier**. Confusion matrices, ROC-AUC curves (0.982), and an interactive 'Retrain Model' button are featured.

- **Natural AI Narration (Tanglish):**

"Next, idhu thaana namma project-oda highlight: **Model Lab & Registry!**
Real-world data science-la oru single model eppovum perfect-ah irukaadhu. Athanaala namma system 5 different algorithms-ah train panni bench test pannuthu:
1. **XGBoost** — Extreme Gradient Boosting. Inga idhu thaana current 'Champion Model', because idhoda inference time just 4 milliseconds, ROC-AUC 0.98!
2. **Random Forest** — Multiple decision trees sernthu vote panra ensemble model.
3. **Logistic Regression & Decision Tree** — Fast baseline models.
Admin-ku venumna oru click-la Vera model-ah 'Promote to Production' pannalam. Zero downtime-la live traffic pudhu model-ku switch aagidum!"

- **Technical Process Being Explained:**

Model Champion/Challenger Framework & Warm Reload: How FraudLens AI uses `joblib` model persistence with in-memory thread-safe hot swapping via `FraudPredictionService.get_instance().set_active_model()`, allowing instantaneous promotion without server restarts.

- **Exact Project UI to Show:**

ModelLabView showing the model comparison table, ROC-AUC badges, and Champion promotion toggles.

- **On-Screen Text:**

MODEL REGISTRY: XGBoost (Champion) • Random Forest • Voting Ensemble

Zero-Downtime Hot Swapping • ROC-AUC: 0.982 • F1-Score: 0.941

- **Cinematic Veo Visual Prompt:**

Holographic benchmark comparison table floating in cyber space, glowing neon bars comparing model accuracies, an interactive neon switch engages and a golden 'Champion' crown animates onto the XGBoost metric card.

- **Transition:** Smooth lateral camera glide into Dataset Health & Drift.

Scene 2.3: Dataset Health & Population Stability Index (PSI)

- **Duration:** 45 Seconds

- **What Appears on Screen:**

The **Dataset Health & Audit View**. Visual meters measure **Data Drift**, **Concept Drift**, and **Feature Integrity**. A gauge for **PSI (Population Stability Index)** shows 0.042 (Clean - No Drift). A feature distribution curve shows baseline vs. inference distribution.

- **Natural AI Narration (Tanglish):**

"Machi, neraya projects-la model train panni vitruvaanga, apram time aaga aaga model accuracy padhukkayila vizhunthidum. Athukku peyar than **Model Drift**! FraudLens AI-la namma **Dataset Health & Drift Monitoring** engine integrate pannirukkom. Inga paaru — **PSI (Population Stability Index)** calculate pannuthu. PSI value 0.1-ku mela poona, training data-kum live transaction patterns-kum periya maatham vanthurukku-nu system automated alert adikkum! Athoda kooda, **Target Leakage** check pannum. Target leakage-na, future information model training-kulla accidental-ah leak aagi fake 100% accuracy kaattama irukiratha verify panra safety mechanism."

- **Technical Process Being Explained:**

Statistical Data Drift & PSI Mathematics: Population Stability Index tracking feature

shifts over time. Alert triggers if PSI exceeds threshold, mandating retraining due to macroeconomic or merchant behavior drift.

- **Exact Project UI to Show:**

DatasetHealthView showing PSI gauges, distribution overlap histograms, and data contract validation cards.

- **On-Screen Text:**

CONTINUOUS DRIFT GOVERNANCE: PSI = 0.042 (Normal < 0.10)

Zero Target Leakage Guaranteed • Automated Data Contract Validation

- **Cinematic Veo Visual Prompt:**

Futuristic digital diagnostics display with animated frequency waves, two luminous curves (baseline vs live) harmoniously overlapping, glowing cybernetic badge reads 'PSI NORMAL // ZERO LEAK'.

- **Transition:** Camera zooms down into the Immutable Audit Logs table.

Scene 2.4: Immutable Audit Trail & ISO-27001 Security

- **Duration:** 35 Seconds

- **What Appears on Screen:**

The **Audit Logs View**. A cryptographic timeline lists every action taken across the entire enterprise: Who logged in, which transaction was blocked, model parameter changes, and API key validations. SHA-256 integrity hashes are displayed for each log row.

- **Natural AI Narration (Tanglish):**

"Final-ah Admin module-la romba mukkiyamaana section: **Audit Logs**.

Banking and FinTech-la accountability than ultimate priority. Yaaru login panna, entha investigator card-ah block pannaanga, model threshold-ah yaaru maathinaanga-nu second-to-second timestamp-oda record aagum.

Every log entry-kum SHA-256 hash cryptographic signature irukku. Even database administrator kooda intha logs-ah tamper panna mudiyaathu!

Idhanaala than namma FraudLens AI ISO-27001 and PCI-DSS compliance standards-ku 100% ready-ah irukku."

- **Technical Process Being Explained:**

Immutable Audit Trail & Forensics: All sensitive operations invoke `AuditLog` database hooks recording `actor_id`, `actor_role`, `action_type`, `payload_hash`, `client_ip`, and `timestamp`, satisfying strict financial auditing mandates.

- **Exact Project UI to Show:**

AuditLogsView with the audit event table, JSON payload inspector modal, and role badges.

- **On-Screen Text:**

IMMUTABLE AUDIT TRAIL // ISO-27001 & PCI-DSS COMPLIANT

SHA-256 Tamper-Proof Event Ledger • Zero Administrative Bypass

- **Cinematic Veo Visual Prompt:**

Extreme macro shot of cryptographic hash keys locking into place like physical metallic tumblers on a glowing holographic ledger, security verification seal animates with emerald energy.

- **Transition:** Screen rotates and transitions into Video 3 (Customer Persona 1: Monisha).

VIDEO 3: Monisha Module (Clean Habitual Baseline — 3% Risk)

Target Duration: ~3 Minutes

Theme: The Safe Habitual Shopper, Zero-Friction Approval & Baseline Match

Scene 3.1: Introducing Monisha & Customer Portal Isolation

- **Duration:** 35 Seconds
- **What Appears on Screen:**

Login screen with 1-click selection of **Monisha (User 1 - Safe Account)**. After clicking, the cyber unlock animation triggers and opens the **Dedicated Customer Dashboard**. Monisha's personalized banking view appears with INR 16,11,121.99 balance, virtual debit card, and green safety badges.
- **Natural AI Narration (Tanglish):**

"Ippo namma 3 realistic customer personas-la first persona-va paapom: **Monisha**. Monisha login pannathum paaru — avangalukku admin tools ethum theriyathu! Internal model lab, drift audit ellam automatic-ah hide aagi, oru ultra-sleek private banking dashboard open aaguthu. Monisha-oda profile enna theriyuma? She is a habitual, trusted shopper. Regular-ah Chennai-la irukura trusted iPhone 15-la irunthu groceries and routine payments pannuvaanga. Avangaloda historical baseline fraud rate just **3%** than!"

- **Technical Process Being Explained:**

Role-Based Access Control (RBAC) & Tenant Isolation: When JWT payload has `role: "customer"` and identity claims for Monisha, `App.jsx` and `Sidebar.jsx` dynamically constrain navigation to customer-relevant endpoints, completely isolating sensitive bank management routes.
- **Exact Project UI to Show:**

CustomerDashboardView with virtual card preview, balance toggle, and recent

transactions list.

- **On-Screen Text:**

PERSONA 1: MONISHA // HABITUAL BENCHMARK

Fraud Baseline: 3% • Trusted Device: iPhone 15 (Chennai) • Flow: ALLOW

- **Cinematic Veo Visual Prompt:**

Elegant light-frosted glass banking portal on a mobile and tablet mockup, glowing emerald badges indicate clean account status, high-end consumer fintech luxury aesthetic.

- **Transition:** Camera smoothly moves to the Pre-Auth Payment Gateway preset.

Scene 3.2: The Live Transaction Demo — Grocery at NovaMart Fresh

- **Duration:** 45 Seconds

- **What Appears on Screen:**

The **Payment Gateway (Pre-Auth)** screen. The scenario preset is clicked: 1.

Monisha: Habitual Grocery (₹1,250). Form fields auto-populate: Amount ₹1,250,

Merchant: *NovaMart Fresh*, Location: *Chennai, IN*, Device: *DEV-MONISHA-IPHONE-15*.

The 'Execute Pre-Auth Evaluation' button is clicked.

- **Natural AI Narration (Tanglish):**

"Demo paapom! Monisha oru ₹1,250 grocery payment NovaMart Fresh-la panraanga.

Naama payment gateway-la 'Execute Pre-Auth' click panrom.

Behind the screen enna aaguthu?

System Monisha-oda past 180 days transaction profile-oda compare pannuthu.

Check 1: Device trusted iPhone 15? Yes!

Check 2: Location usual Chennai IP address? Yes!

Check 3: Amount routine grocery purchase band-la irukka? Yes!

Model evaluate panni instant-ah **Probability: 2.1%** and **Risk Score: 14/100 (LOW)** assign pannuthu!"

- **Technical Process Being Explained:**

Behavioral Profile Matching Service: How `behavior_profile_service.py` computes z-scores for amount deviation, validates known trusted device identifiers, and outputs low baseline risk.

- **Exact Project UI to Show:**

PaymentView showing instant calculation and green celebration animation.

- **On-Screen Text:**

TRANSACTION #TX-MON-8912 // ₹1,250.00

Fraud Probability: 2.1% • Operational Risk: 14/100 • Verdict: AUTO-ALLOW

- **Cinematic Veo Visual Prompt:**

Futuristic transaction validation HUD: biometric fingerprint match lights up in soft emerald green, checkmarks ripple across transaction attributes with zero resistance.

- **Transition:** Quick cut to the instant checkout completion status.

Scene 3.3: Zero Friction & Explainable Green Telemetry

- **Duration:** 40 Seconds

- **What Appears on Screen:**

The transaction status confirms: **ALLOW (Auto-Approved)**. The SHAP explanation card displays: `device_trust_score (-0.32)`, `merchant_familiarity (-0.28)`, `geo_consistency (-0.19)`. Negative SHAP values clearly prove why the transaction was approved with zero user friction (no OTP prompt).

- **Natural AI Narration (Tanglish):**

"Result paarum: **ALLOW (Auto-Approved)**!"

Monisha-kku entha OTP popup-um varaama, payment single-second-la complete aagidum. Idhukku peyar than **Frictionless Consumer Experience**.

SHAP waterfall chart-la paatha, ellame green bars left side push aaguthu: Trusted device -0.32, Regular merchant -0.28.

Genuine customer-ku seamless payment tharanum, athoda bank security-um compromise aaga koodathu. Adha Monisha module 100% demonstrate pannuthu!"

- **Technical Process Being Explained:**

Frictionless Authentication Thresholds: Transactions with Risk Score < 30 bypass Step-Up challenges (PSD2 / RBI SCA exemption criteria for trusted low-value merchant patterns).

- **Exact Project UI to Show:**

The green explanation card inside PaymentView and balance deduction update in CustomerDashboardView.

- **On-Screen Text:**

FRICTIONLESS SECURITY: Zero User Interruption

RBI/PSD2 SCA Exemption Applied • Negative SHAP Pull To Safe Zone

- **Cinematic Veo Visual Prompt:**

Close-up shot of a glowing green holographic shield pulsing softly over a virtual payment card, digital receipt unfolds seamlessly in sleek glassmorphism.

- **Transition:** Scene darkens with an amber warning glow transitioning into Video 4 (Mohana).

VIDEO 4: Mohana Module (Suspicious Velocity — 12% Risk & Step-Up OTP)

Target Duration: ~3 Minutes

Theme: Rapid Micro-Burst Velocity, Unfamiliar Region & Adaptive Step-Up Verification

Scene 4.1: Introducing Mohana & The Unfamiliar Pattern

- **Duration:** 35 Seconds
- **What Appears on Screen:**
Login as **Mohana (User 2 - Step-Up OTP)**. The customer portal displays an amber alert banner on the dashboard. The profile details show a baseline fraud risk of **12%**. A notification warns of an active Step-Up challenge on a pending transaction.
- **Natural AI Narration (Tanglish):**

"Next, namma second persona-va paapom: **Mohana**.
Mohana-oda baseline risk **12%** — moderate review band.
Monisha maari illama, Mohana case-la oru unusual situation create aaguthu.
Mohana sudden-ah oru unfamiliar device-la irunthu, different network-la login panni, short duration-la rapid-ah multiple transactions initiate panraanga.
Intha scenario-va thaana namma **Velocity Burst Anomaly**-nu solrom!"

- **Technical Process Being Explained:**
Velocity Anomaly & The Moderate Risk Band: When transactions fall into the 30 <= Risk Score < 70 zone, automatic approval is too risky, but hard-blocking will frustrate a legitimate customer. Adaptive Step-Up OTP is triggered.
- **Exact Project UI to Show:**
CustomerDashboardView with the amber theme and active security challenge indicator.
- **On-Screen Text:**

PERSONA 2: MOHANA // SUSPICIOUS VELOCITY BAND

Fraud Baseline: 12% • Unfamiliar Device IP • Action: STEP-UP OTP CHALLENGE

- **Cinematic Veo Visual Prompt:**

Warm amber-lit financial dashboard displaying fluctuating pulse lines, an amber security beacon rotates smoothly indicating an active verification hold.

- **Transition:** Push-in to the Payment Gateway simulation of the micro-burst transaction.

Scene 4.2: Payment Execution & Velocity Spike Detection

- **Duration:** 45 Seconds

- **What Appears on Screen:**

In PaymentView, preset **2. Mohana: Rapid Micro-Burst Sequence** is triggered. Amount: ₹890, 9 transactions in the last 45 seconds. The real-time evaluation immediately flags the velocity cluster. Risk Score jumps to **68/100 (MEDIUM / OTP)**.

- **Natural AI Narration (Tanglish):**

"Watch this live demo! Mohana account-la irunthu ₹890-kku transaction execute panrom. But system immediate-ah alert aaguthu: 'Wait! Last 45 seconds-la idhu 9th transaction attempt!'"

Feature engineer calculate pannuthu: **trans_velocity_1h** standard threshold-a vida +300% athigama irukku!

Plus, used IP address unfamiliar ISP-la irukku.

XGBoost model fraud probability-ah **64.2%**-nu calculate pannuthu, and operational risk score **68/100**-nu jump aaguthu!

Hard block panna poroma? No! Customer-a verify panna Step-Up OTP challenge trigger panrom!"

- **Technical Process Being Explained:**

Sliding Window Velocity Calculation: Frequency count of transactions over rolling time frames. High frequency triggers Rule Engine velocity challenge.

- **Exact Project UI to Show:**

The transactional cluster bar chart inside EvidenceDashboard and the risk gauge hitting 68/100.

- **On-Screen Text:**

BURST DETECTED: 9 Txns / 45 Seconds

Risk Score: 68/100 • SHAP Driver: trans_velocity_1h (+0.36)

- **Cinematic Veo Visual Prompt:**

Vertical bar graph surges upward with illuminated amber bars spiking into the danger threshold, numeric counter counts rapidly from 1 to 9.

- **Transition:** Cut to the interactive Mobile Security Approval Modal.

Scene 4.3: Real-Time Mobile Security Approval Modal

- **Duration:** 40 Seconds

- **What Appears on Screen:**

The **MobileSecurityApprovalModal** pops up on screen representing Mohana's phone. A secure 6-digit OTP prompt and biometric approval button appear. The user enters **849201** and clicks **Authorize**. The transaction transitions to APPROVED with security logged.

- **Natural AI Narration (Tanglish):**

"Screen-la paaru — namma custom **Mobile Security Approval Modal** open aaguthu! Real world-la customer phone-ku instant push notification pogum: 'Intha ₹890 transaction neenga than panreengala?' Customer avanga 6-digit OTP enter panni approve pannathum, request backend-la **/api/v1/approvals/verify** endpoint-ku pogum. Cryptographic validation mudinjathum, transaction instant-ah approve aagi funds release aagum! Idhanaala fraud-um thadukapaduthu, genuine user-oda payment-um reject aagala. Perfect balance!"

- **Technical Process Being Explained:**

Two-Factor Cryptographic Step-Up Resolution: The approval session stores a cryptographic nonce with a 180-second TTL. Upon verified input, the state flips from PENDING to AUTHORIZED.

- **Exact Project UI to Show:**

MobileSecurityApprovalModal with live timer countdown, OTP input boxes, and approval confirmation.

- **On-Screen Text:**

CRYPTOGRAPHIC STEP-UP CHALLENGE // 180s TTL

Status: AUTHENTICATED BY CUSTOMER • Risk Cleared Safely

- **Cinematic Veo Visual Prompt:**

Sleek 3D smartphone holographic render displaying an animated biometric fingerprint ring glowing in amber, shifting to luminous teal upon authorized touch.

- **Transition:** Screen turns to deep crimson with emergency sirens pulsing into Video 5 (Sowmiya).
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VIDEO 5: Sowmiya Module (Botnet ATO Attack — 26% Risk, Block & SAR)

Target Duration: ~3 Minutes 30 Seconds

Theme: Impossible Geo-Leap Travel, Botnet Account Takeover (ATO) & FinCEN SAR Filing

Scene 5.1: The Threat — Botnet Credential Stuffing on Sowmiya

- **Duration:** 35 Seconds
- **What Appears on Screen:**

Login portal selecting **Sowmiya (User 3 - ATO Attack)**. The screen displays an alert indicator in dark crimson. Dashboard metrics reflect a **26% baseline fraud attack profile**. The system alerts that account credentials were leaked in an external breach and are being targeted by an automated botnet proxy network.
- **Natural AI Narration (Tanglish):**

"Ippo namma paaka porathu FraudLens AI-oda ultimate stress test: **Sowmiya Module — The Botnet Account Takeover (ATO) Attack**.

Sowmiya oru high-risk baseline account (26%). External data breach-la ivanga password leak aayirukku.

Dark web-la irukura automated botnet syndicates Sowmiya account-ah target pannu, credential stuffing valiya ulla enter aaga try panraanga.

But FraudLens AI-oda multi-layered defense irukkurathaala, hacker-oda game inga mudiypoguthu!"

- **Technical Process Being Explained:**

Account Takeover (ATO) Threat Mechanics: Automated botnets use headless browser clusters rotating through residential proxy networks to drain compromised user accounts.
- **Exact Project UI to Show:**

CustomerDashboardView rendering in high-alert crimson with active security lock notices.

- **On-Screen Text:**

PERSONA 3: SOWMIYA // BOTNET ATO ATTACK

Attack Vector: Credential Stuffing • Threat Tier: CRITICAL • Defense: HARD BLOCK

- **Cinematic Veo Visual Prompt:**

Ominous digital cyber map showing distributed automated botnet attack vectors targeting a central bank account node, shielded by a glowing hexagonal firewall barrier.

- **Transition:** Cut to the Payment Gateway executing the impossible geo-leap transaction.

Scene 5.2: The Impossible Geo-Leap Anomaly (Bucharest, Romania)

- **Duration:** 45 Seconds

- **What Appears on Screen:**

The Payment Gateway triggers **3. Sowmiya: Bucharest Impossible Geo-Travel (\$3,850.00)**. Just 18 minutes ago, a transaction occurred in San Francisco. Now, a massive \$3,850 transaction is attempted from a datacenter IP in **Bucharest, Romania**. The **Geo-Leap Interactive Vector Map** in EvidenceDashboard displays a glowing red arc linking SF to Bucharest with velocity: **28,066 km/h (Physically Impossible)**!

- **Natural AI Narration (Tanglish):**

"Intha attack epdi execute aaguthu-nu paaru!
Just 18 minutes munnadi, Sowmiya San Francisco-la oru legitimate coffee purchase pannanga.
Exactly 18 minutes apram, Romania-oda Bucharest IP-la irunthu sudden-ah \$3,850 international transfer attempt aaguthu!
18 minutes-la oru human 8,420 km travel panna mudiyuma? That means flight speed 28,000 km per hour! It is **Physically Impossible**!
Namma Geo-Leap telemetry engine idhai instant-ah flag pannuthu: 'Datacenter VPN proxy confirmed, impossible velocity detected!'"

- **Technical Process Being Explained:**

Haversine Great-Circle Distance & Velocity Anomaly: Calculating geographical coordinates against timestamp deltas to identify speeds exceeding physical human travel limits.

- **Exact Project UI to Show:**

EvidenceDashboard with the interactive SVG World Map showing the arched red flight trajectory and Bucharest VPN tag.

- **On-Screen Text:**

GEO-LEAP VELOCITY: 28,066 km/h // PHYSICALLY IMPOSSIBLE

San Francisco (18m ago) → Bucharest, Romania (Attempted Now)

- **Cinematic Veo Visual Prompt:**

Glowing 3D world globe showing an illuminated trajectory arc soaring across the Atlantic Ocean from California to Eastern Europe, pulsing crimson warning labels flash 'IMPOSSIBLE VELOCITY'.

- **Transition:** Dramatic trigger of the automated hard block and FinCEN SAR draft card.

Scene 5.3: Automated Card Freeze & Regulatory FinCEN SAR Filing

- **Duration:** 50 Seconds

- **What Appears on Screen:**

The transaction is instantaneously marked **BLOCKED**. Fraud Probability hits **91.8%**, Risk Score hits **94/100 (CRITICAL)**. The account card automatically flips to FROZEN. Simultaneously, the **SarReportDraftCard** renders a regulatory **FinCEN Suspicious Activity Report (SAR)** ready for compliance sign-off.

- **Natural AI Narration (Tanglish):**

"System immediate-ah action execute pannuthu: **PAYMENT BLOCKED!** Zero compromise! Risk Score hits **94/100!** Automatic-ah Sowmiya-oda virtual card freeze aagidum, so hacker-aala adutha single rupee kooda edukka mudiyathu. Athoda mudinjithu-nu ninaikareengala? No! FinTech regulations padi, international cyber attack nadantha bank government-ku report pannanum. Namma system automated-ah oru complete **FinCEN SAR (Suspicious Activity Report)** draft generate pannidum! Hacker-oda suspect IP, impossible velocity evidence, TreeSHAP scores ellame formatted narrative-ah ready-ah irukkum. Compliance officer just review panni submit panna pothum!"

- **Technical Process Being Explained:**

Automated Compliance & FinCEN Suspicious Activity Reporting (SAR): Under Bank Secrecy Act rules, illicit attempts exceeding statutory limits involving cyber-identity theft trigger automated SAR drafting with detailed audit telemetry.

- **Exact Project UI to Show:**

SarReportDraftCard with formatted narrative, copy buttons, and card freeze toggle in CustomerDashboardView.

- **On-Screen Text:**

CRITICAL ACTION: CARD FROZEN • FUNDS PROTECTED: \$3,850.00

Automated FinCEN SAR Draft Generated // Case File #CASE-8817

- **Cinematic Veo Visual Prompt:**

A glowing red digital lock clamps down tightly onto a digital credit card, a formal compliance document unfolds with holographic forensic stamps and verified signatures.

- **Transition:** Camera pulls out into a high-tech glass studio setting up Video 6 (Chatbot).



VIDEO 6: Chatbot Module (AI Intelligence Workspace & Multi-LLM Orchestrator)

Target Duration: ~3 Minutes 30 Seconds

Theme: Light Frosted Glass Workspace, Holographic AI Orb, Multi-LLM Orchestrator & Forensic Synthesis

Scene 6.1: The Intelligence Workspace (1:1 Reference Video Match)

- **Duration:** 35 Seconds
- **What Appears on Screen:**

The complete **FraudLens AI Intelligence Workspace** rendering in the exact **Light Frosted Glass Theme** from `gemini_generated_video_2ce150b8.mp4`. The top pill header displays `Sarah J. / Analyst`, bold `FRAUDLENS AI`, and frosted model buttons `[AUTO]` `[GEMINI]` `[GROK]` `[MISTRAL]`. The 3D Holographic AI Orb breathes softly in the center.
- **Natural AI Narration (Tanglish):**

"Final-ah, namma showcase panra crown jewel: **FraudLens AI Intelligence Workspace & Chatbot!**

Screen-ah paaru — exact reference video maari oru ultra-premium **Light Frosted Glass** visual design!

Top-la namma analyst badge 'Sarah J.', center-la sleek pill capsule with multi-model selectors: Auto, Gemini, Grok, and Mistral.

Center-la oru bioluminescent **3D Holographic AI Orb** float aaguthu. Intha orb simply decoration kedayathu — backend-la AI enna yosikkuthu-nu athoda internal particles and orbital rings live-ah indicate pannum!"

- **Technical Process Being Explained:**

Holographic Canvas State Engine: AIOrb uses HTML5 Canvas 2D math calculating counter-rotating harmonic ellipses with volumetric particle dynamics reflecting 7 discrete inference states (idle, listening, searching, analysing, deep_reasoning, cross_validating, ready).

- **Exact Project UI to Show:**

AilInvestigationCommandCenter showing the 3-panel layout, top pill header, and floating AI Orb.

- **On-Screen Text:**

FRAUDLENS AI INTELLIGENCE WORKSPACE // MULTI-LLM ARCHITECTURE

Frosted Glassmorphism • Bioluminescent Holographic Orb • Sub-10ms UX

- **Cinematic Veo Visual Prompt:**

Photorealistic cinematic shot of a light frosted glass cyber investigation studio, bright high-tech aesthetic, in the center a fluid iridescent chrome and pearl holographic orb floats with glowing cyan and soft ivory light rings, 8k resolution.

- **Transition:** Focus snaps onto the chat input bar entering the prompt.

Scene 6.2: The Forensic Query & Live Radar Scanning Beam

- **Duration:** 45 Seconds

- **What Appears on Screen:**

The user enters the prompt from the reference video: "Analyze transactions for Account #8812 for the last 14 days". The send button is clicked. Immediately, the **AI Orb shifts to a radial radar sweep beam**. The status badge cycles in real-time: 

Searching Evidence Vectors... -> ⚡ Scanning Global Risk Nodes... -> 
Analysing TreeSHAP Drivers... -> ✨ Synthesizing Defense Report...

- **Natural AI Narration (Tanglish):**

"Ippo namma exact video prompt-ah enter panrom: 'Analyze transactions for Account #8812 for the last 14 days'.
Send click pannathum paaru — AI Orb immediately radar sweep mode-ku shift aaguthu!
Screen-la live stage sequencer animate aaguthu:
Searching -> Scanning -> Analysing -> Synthesizing!
Background-la namma **Multi-LLM Orchestrator** work aaguthu.
Normal chatbot mathiri idhu verum text generator kedayathu. Idhu database-la irukkira live transaction records, TreeSHAP scores, and global blacklist IP-ah gather panni oru complete forensic dossier ready pannuthu!"

- **Technical Process Being Explained:**

Autonomous Multi-LLM Orchestration & Evidence Grounding: [orchestrator.py](#)

injects live forensic domain context from `evidence_service.py`, verifying facts against actual database records to eliminate LLM hallucinations before generation.

- **Exact Project UI to Show:**

The animated scanning badge and AI Orb radar beam inside AilInvestigationCommandCenter.

- **On-Screen Text:**

AUTONOMOUS ORCHESTRATION: Searching → Scanning → Analysing → Synthesizing

Zero Hallucination Grounding • Multi-Agent Evidence Extraction

- **Cinematic Veo Visual Prompt:**

Macro close-up of a radial radar beam sweeping across the iridescent sphere, data points highlight in sapphire blue, sleek glowing text indicators animate smoothly in sequence.

- **Transition:** The AI response card and evidence dashboard light up simultaneously.

Scene 6.3: Dual Synthesis — Gemini Primary + Grok Forensic Challenge

- **Duration:** 50 Seconds

- **What Appears on Screen:**

The frosted AI response card appears: "Analysis of 328 transactions completed. Key findings include multiple high-velocity small-value transactions across Latin America locations..." with a live link: `↳ Potential link identified: Case #4501 share same IP subnet`. The right-side **Evidence Dashboard** syncs up, displaying the LATAM world map, transaction clusters, and the 5-node interconnected account graph.

- **Natural AI Narration (Tanglish):**

"Response vanthuruchu, paaru!

'Analysis of 328 transactions completed. Key findings include multiple high-velocity small-value transactions across Latin America locations!'

Inga thaana namma **Dual-LLM Synthesis** masterstroke irukku:

Primary analysis-a generate panrathu **Google Gemini 3.6 Flash**.

Athey nerathula, **xAI Grok** independent forensic auditor-ah act panni, 'Case #4501 share the same IP subnet'-nu cross-challenge pannum!

Rendu models-oda intelligence-um combine aagi, no blind consensus illama, verified final report UI-la display aaguthu.

Right side-la paatha, connected account network graph-la Account 1, 2, 3, 4, 5 epdi correlate aaguthu-nu interactive-ah paarkalam!"

- **Technical Process Being Explained:**

Dual-Synthesis Consensus Engine: Combining primary analysis with independent forensic challenge to guarantee verified adversarial validation without blind consensus.

- **Exact Project UI to Show:**

The AI message card in AiInvestigationCommandCenter and the connected 5-node graph in EvidenceDashboard.

- **On-Screen Text:**

DUAL-SYNTHESIS: Google Gemini (Primary) + xAI Grok (Adversarial Audit)

Mule Network Correlated: 5 Interconnected Nodes Identified

- **Cinematic Veo Visual Prompt:**

Sleek frosted glass data cards assemble gracefully on screen, glowing connections draw lines between node circles #8812 and #4501, ambient studio light reflections.

- **Transition:** Camera glides down to the action trigger pills.

Scene 6.4: One-Click Investigator Action & Grand Project Conclusion

- **Duration:** 35 Seconds

- **What Appears on Screen:**

The investigator clicks the **[Draft Report]** and **[Block Account]** pills below the input bar. A FinCEN SAR draft appears, the account status locks, and all six modules of FraudLens AI animate into a synchronized 3D ecosystem grid. The FraudLens AI crest glows proudly in the center.

- **Natural AI Narration (Tanglish):**

"Bottom-la paaru — Investigator single click-la **[Draft Report]**, **[Block Account]**, **[Escalate]** action execute பண்ணலாம்.

So machi, total-ah summary paatha:

Historical and Live transactions-a capture panrom...

5 Machine Learning models vachu 4 milliseconds-la evaluate panrom...

TreeSHAP valiya transparent explainability tharom...

Monisha, Mohana, Sowmiya-nu complete role isolation ensure panrom...
And multi-LLM AI Copilot vachu complex financial crime syndicates-a burst panrom!
This is not just a project — this is the **Future of Explainable FinTech AI!**"

- **Technical Process Being Explained:**

End-to-End Platform Cohesion: How every tier (Frontend -> REST APIs -> ML Pipelines -> Explainability -> Autonomous LLM Orchestration) connects into a single enterprise-grade ecosystem.

- **Exact Project UI to Show:**

The complete synchronized view with the quick action pills in AilInvestigationCommandCenter.

- **On-Screen Text:**

FRAUDLENS AI // THE FUTURE OF EXPLAINABLE FINTECH INTELLIGENCE

Zero-Downtime • Sub-10ms Inference • Multi-LLM Orchestration • 100% Verified

- **Cinematic Veo Visual Prompt:**

Grand cinematic pull-back shot: all six floating glass modules converge into an interlocking 3D holographic sphere with the FraudLens AI shield glowing powerfully at the center, fiber-optic light pulses through the network, clean professional cyber-fintech lighting, fade to black.

- **Transition:** Final cinematic fade-out with audio confirmation chime.